

Algorithm Analysis & Complexity Theory

Sample Exam: Dynamic Programming

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Difficulty Levels: Foundation, Conceptual, Application, Advanced

Question 1 [Difficulty: Conceptual]

Explain the optimal substructure property of the longest common subsequence (LCS) problem.

Solution:

Optimal Substructure of LCS:

Let $X = x_1x_2 \dots x_n$ and $Y = y_1y_2 \dots y_m$ be two sequences

Let $L[i, j]$ = length of LCS of $X[1..i]$ and $Y[1..j]$

Property:

If $x_i = y_j$:

$$L[i, j] = L[i-1, j-1] + 1$$

If $x_i \neq y_j$:

$$L[i, j] = \max(L[i-1, j], L[i, j-1])$$

Proof that this is optimal:

Case 1: If $x_i = y_j$ and both are in the LCS

- The LCS must end with $x_i = y_j$
- Remove this character: remaining is LCS of $X[1..i-1]$ and $Y[1..j-1]$
- If it wasn't optimal, we could construct a longer LCS - contradiction

Case 2: If $x_i \neq y_j$

- At least one is not in the LCS
- LCS is either LCS of $X[1..i-1], Y[1..j]$ or $X[1..i], Y[1..j-1]$
- We take the maximum of both

This recursive structure leads to the DP solution ■